Response to: Request for Information – Health Technology Ecosystem
Source: US Department of Health and Human Services (HHS), Centers for Medicare and Medicaid Services (CMS)
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Note the RFI Summary lead-in, "Effective and responsible adoption of technology can empower patients to make better decisions for their health and well-being". We believe this Administration is uniquely qualified to turn the tables on an HIT infrastructure currently subservient to everything big: big government and big institutions, including big providers, big health plans, big vendors, big pharma...

"Patient-centric" and "patient empowerment" are contending buzzwords but are entirely peripheral considerations to many of the most highly-promoted HIT infrastructure "solutions". Individual clinicians have also been brushed aside, with little attention to their pleas for focus on usability, efficacy and burden reduction. Deserving they are, yet bearing little recognition, ever relegated to the role of small cogs in a big wheel.

Although we didn't have time to fully address every RFI question we worked to examine muddles and mania in current HIT infrastructure and to offer worthy suggestions in our quest for "effective and responsible adoption of technology [to] empower patients".

PC-1 – What health management or care navigation apps would help you understand and manage your (or your loved ones) health needs, as well as the actions you should take?

[See responses to PC-1c below]

PC-1a – What are the top things you would like to be able to do for your or your loved ones' health that can be enabled by digital health products?

[See responses to PC-1c below]

PC-1b – If you had a personal assistant to support your health needs, what are the top things you would ask them to help with? In your response, please consider tasks that could be supported or facilitated by software solutions in the future?

- Review and manage:
  - IDs, identity parameters
  - Family history
  - Genetics
  - Encounter history
  - Appointment management: scheduling and re-scheduling
  - Messages, interactions with providers and caregivers
  - Medication management: prescribed and over the counter, reconciliation, re-fills, past and present
  - Alerts, reminders and notifications
  - Advance directives for healthcare
  - Consents
    - For healthcare procedures and interventions, past and present
    - For health/healthcare information disclosure, past and present
    - For AI:
      - For workup, diagnosis and treatment
      - For machine learning, utilizing individual health/healthcare information
  - Claims and payment, past and present

- Review, correct and annotate:
  - Demographics, current contact information
  - Next of kin
  - Insurance, policy and plan coverage, subscribers, guarantors, claims and payments, past and present
  - History and physical findings
  - Vital signs
  - Problems, conditions, signs, symptoms, past and present
  - Immunizations, past, present, future
  - Allergies and intolerances
  - Assessments, progress notes, past and present
  - Care plans, past and present
  - Therapies
  - Orders and prescriptions, past and present
  - Diagnostic results, past and present
  - Social determinants of health
  - Referrals, consultations: plans and findings
  - Medical devices, durable medical equipment, implants, routine supplies, past and present
  - Procedures, interventions, past and present
  - Nutrition and diet
  - Transportation
- Review chronology (personal timeline) of health and health care - interactions and events:
  - Retrospective (past)
  - Concurrent (present)
  - Prospective (anticipated, future)
- Access directory of personal providers and caregivers: location, contact details, past and present

PC-2 – Do you have easy access to your own and all your loved ones' health information in one location (for example, in a single patient portal or another software system)?

- Only if there is a single provider for all care.
- Otherwise each provider has a separate portal - which could amount to a dozen or more per individual.
- Each portal has its own variant way to access, navigate, display and alert/notify.
- The need is for independent PHR and/or Health Record Bank accounts (not EHR portals), where an individual can direct their health information be sent, e.g., after each encounter.
- This is the centered model = one patient one record - single source of truth.
- Obviates the need for schemes such as TEFCA where individual health information is scattered all over the place in dissonant fragments.

PC-2a – If so, what are some examples of benefits it has provided?

- All health information for an individual is directed to a single logical location and dispersed, as needed and authorized, from that same location.
- Provides a single point of access to, and full integration of, an individual's health information.
- Enables individual, not institutional, control of personal health information.
- Encourages providers to talk *with* the individual, not *about* the individual.

PC-2b – If not, in what contexts or for what workflows would it be most valuable to use one portal or system to access all such health information?

- Single source for:
  - Demographics, current contact information, next of kin
  - Insurance, policy and plan coverage, subscribers, guarantors, claims and payments, past and present
  - Personal providers and caregivers: directory (location, contact information), past and present
- Curated (reconciled, integrated) lists:
  - Medications, prescribed and over the counter
  - Immunizations

    - Problems, conditions
    - Allergies, intolerances
    - Procedures, interventions
- Consolidated chronology, personal timeline of health and health care - interactions and events:
    - Retrospective (past)
    - Concurrent (present)
    - Prospective (anticipated, future), include appointments
- Individuals "own" a copy of their health record as instantiated in their PHR/HRB account
- Individuals can directly interact with their providers and participate in coordination of their healthcare
- Individual health/healthcare activities can be both monitored and prompted (guided)
- Individuals can directly keep an eye on their health record for:
    - Anticipated (or other) updates
    - Accuracy, consistency, completeness
    - Deposits (data in) and withdrawals (data out)
- Individuals can annotate and request corrections to health record content

[For further discussion, see response to PC-10 below]

PC-2c – Were there particular data types, such as x-rays or specific test results, that were unavailable? What are the obstacles to accessing your own or your loved ones' complete health information electronically and using it for managing health conditions or finding the best care (for example, limitations in functionality, user friendliness, or access to basic technology infrastructure)?

[See response to PC-4 below]

PC-3 – Are you aware of health management, care navigation, or personal health record apps that would be useful to Medicare beneficiaries and their caregivers?

[See responses to PC-1b and PC-2b above and PC-10 below]

PC-4 – What features are missing from apps you use or that you are aware of today?

- Apps are largely missing the ability to meet these universal challenges – to ensure trustworthiness:
    - You don't know what you don't know. Are health data/records complete? What might be missing?
    - You have little basis to trust health data/record content
    - You often don't know basic provenance for health data/record content, including:
        - Who: author, organization
        - What: related event or action taken
        - When: action was taken and data captured
        - Where: physical location, device ID, network address
        - Why: rationale, purpose of capture

[See data qualities in response to PC-11 below]

PC-4a – What apps should exist but do not yet? Why do you believe they do not exist yet?

- Apps that cross-validate AI results.
- Apps that cross compare providers and services: credentials, qualifications, rating/satisfaction, accessibility, costs...

PC-4b – What set of workflows do you believe CMS is uniquely positioned to offer?

[No response due to time constraints]

PC-5 – What can CMS and its partners do to encourage patient and caregiver interest in these digital health products?

- Set aside the federal EHR/HIT Incentive Program in all its forms, including EHR/HIT system certification
  - It has served its purpose and is past its prime: >98% inpatient adoption of comprehensive EHR systems, >90% ambulatory care adoption
- Return to private sector, market-driven, completely voluntary EHR/HIT system certification
  - Include functional requirements developed by *actual EHR system end users* (specifically clinicians, caregivers, patients and others) and tuned to their workflows, navigation, inquiry, access and display needs
  - Model after the Certification Commission for HIT (CCHIT), circa 2004-2012
  - Reference industry consensus standards, such as:
    - ISO/HL7 10781 Electronic Health Record System Functional Model (now Release 2.1)
    - ISO/HL7 16527 Personal Health Record System Functional Model (now Release 2)

PC-5a – What role, if any, should CMS have in reviewing or approving digital health products on the basis of their efficacy, quality or impact or both on health outcomes (not approving in the sense of a coverage determination)? What criteria should be used if there is a review process? What technology solutions, policy changes, or program design changes can increase patient and caregiver adoption of digital health products (for example, enhancements to data access, reimbursement adjustments, or new beneficiary communications)?

First consider statutory obligations...

HITECH Act (2009), Section 3001(c)(3)(A): "The National Coordinator [ONC] shall, in consultation with other appropriate Federal agencies (including the National Institute of Standards and Technology), update the Federal Health IT Strategic Plan... to include specific objectives, milestones, and metrics with respect to the following:
"(i) The electronic exchange and use of health information and the enterprise integration of such information.
"(ii) The utilization of an electronic health record for each person in the United States by 2014…"

- As described in the HITECH Act (sub-bullet (ii) above), it is incumbent on CMS/ASTP/ONC to develop "specific objectives, milestones, and metrics with respect to… the utilization of an electronic health record for each person in the United States".
- HITECH does not say "an electronic health record for each provider", yet a substantial fortune (>$45B USD) has been spent in incentives for provider-based EHR systems, thus ensuring that "each person" in the United States had a dozen or more separate electronic health records. This is equivalent to, and perhaps worse than, having none.
- The assumption seems to be that person-centeredness will appear as an emergent phenomenon, without the need for any actual constituent part that unambiguously represents the person in the overall picture.

21st Century Cures Act (2016), Section 4006(a), Encouraging Patient Access to Health Information

"The Secretary shall use existing authorities and take additional actions... with the goal of offering patients access to their electronic health information in a single, longitudinal format that is easy to understand, secure, and may be updated automatically..."

- Over extended time CMS/ASTP/ONC has done little to establish or promote a person-centered electronic health record, inclusive of health information over time from all providers. This was the point of HITECH and later the 21st Century Cures Act. Taking better care of providers and their records – if that's what's been accomplished – has a tenuous connection at best to taking better care of patients.

PC-5b – What changes would enable timely access to high quality CMS and provider generated data on patients?

- Important to define "high quality data"... What are its characteristics, how is it measured, how is it ensured?
- Data quality, or lack thereof, plagues all end uses and all end users of health information today. Everyone wants to rely on digitized data but we have vanishingly little upon which to base trust/assurance – other than the virtue that it was captured, managed and rendered by software.

[See in depth discussion of "data quality" in response to PC-11 and PC-11a/b below.]

PC-6 – What features are most important to make digital health products accessible and easy to use for Medicare beneficiaries and caregivers, particularly those with limited prior experience using digital tools and services?

[See response to PC-5a above]

- Per HITECH: "The utilization of an electronic health record for each person..."
- Per 21st Century Cures Act: "...with the goal of offering patients access to their electronic health information in a single, longitudinal format that is easy to understand, secure, and may be updated automatically..."
- One patient, one record = single source of truth for each patient.

PC-7 – If CMS were to collect real-world data on digital health products' impact on health outcomes and related costs once they are released into the market, what would be the best means of doing so?

[No response due to time constraints]

PC-8 – In your experience, what health data is readily available and valuable to patients or their caregivers or both?

[See responses to PC-1b and PC-2b above]

PC-8a – What data is valuable, but hard for patients and caregivers, or app developers and other technical vendors, to access for appropriate and valuable use (for example, claims data, clinical data, encounter notes, operative reports, appointment schedules, prices)?

- Valuable data is that which is high quality and is verifiably so.

PC-8b – What are specific sources, other than claims and clinical data, that would be of highest value, and why?

[See responses to PC-8a above and PC-11 and its subsections below]

PC-8c – What specific opportunities and challenges exist to improve accessibility, interoperability and integration of clinical data from different sources to enable more meaningful clinical research and generation of actionable evidence?

- Opportunities = one patient one record = single source of truth for each patient.
- Enabled by independent PHR and Health Record Bank accounts.
- As set forth in functional requirements in ISO/HL7 16527 PHR System Functional Model Release 2.0.1.

[See in depth discussion in response to PC-10 below.]

PC-9 – Given that the Blue Button 2.0 API only includes basic patient demographic, Medicare coverage, and claims data (Part A, B, D), what additional CMS data sources do developers view as most valuable for inclusion in the API to enable more useful digital products for patients and caretakers?

- Based on proper design, implementation of one patient one record – single source of truth for each patient – is sufficient to supplant Blue Button.

PC-9a – What difficulties are there in accessing or utilizing these data sources today?

- Tailored specifically to data submitted to substantiate claims for payment, i.e., that which is billable
- Fail to represent true clinical state and health condition, including full detail of activities to support individual health and provide healthcare services

PC-9b – What suggestions do you have to improve the Blue Button 2.0 API experience?

[See response to PC-9 above]

PC-9c – Is there non-CMS data that should be included in the API?

[See responses to PC-1b, PC-2a/b, PC-5, PC-9 above and PC-10 below]

PC-10 – How is the Trusted Exchange Framework and Common Agreement™ (TEFCA™) currently helping to advance patient access to health information in the real world?

"It is hard to escape the conclusion that the antiquated architecture of TEFCA, the baroque construction of 'Individual Access Services' and the constant failure to use modern APIs and data formats suggests that TEFCA is mostly about preventing digital competition for incumbents and digital choice for patients. A lot of work to prevent our smartphones providing us with tools for health at a reasonable price! Of note HIPAA requires either explicit patient consent or a signed BAA agreement to share PHI. TEFCA's 'permitted uses' and various faux HIPAA constructs have never been through the Administrative Procedure Act's Notice of Public Rule-Making = TEFCA data transfers without explicit patient permission or a demonstrable 1:1 BAA are almost certainly HIPAA breaches. Availability of 'after the fact' searches of audit logs does not remedy the breach." – Don Rucker MD via LinkedIn

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The Stated Objective...
"The vision we seek to achieve is a system where *individuals are at the center of their care* and where providers have the ability to securely access and use health information from different sources. A system where an individual's health information is not limited to what is stored in electronic health records (EHRs), but includes information from many different sources (including technologies that individuals use every day) and provides a longitudinal picture of their health." – TEFCA Introduction, 5 January 2018 [italics added]

TEFCA – The Scatter Model

- TEFCA is a prime example of how the US Government has demanded, effectively enabled and enforced the separation of individuals and their health information.
- TEFCA ensures individual health information scattered across multiple sources
  - As diverse and disparate fragments
- TEFCA builds on a HIN of HINs. The very postulation of such an overarching framework/agreement instantiates the combinatorial explosion of complexities between myriad institutional EHR/HIT systems each with distinct conventions, policies and governance and methods, evidenced as disparate:
  - Individual IDs, identity parameters
  - Event, activity and healthcare procedure definitions
  - Data definitions: structured and unstructured, data types, formats, precisions, content and reference ranges, units of measure, code/value sets
  - Content/context patterns and binding
  - Metadata and provenance
  - Mechanisms for authorization (permissions) and consent management
  - Mechanisms for user authentication and access control
  - Mechanisms for accountability
  - Mechanisms for audit: events/triggers, logs, log content

The Vanishing Individual

- HITECH, the 21st Century Cures Act and even TEFCA claim to aspire for each individual to have their own record and be at the center of their own care
- With TEFCA...
  - There is nowhere to point to that is that center.
  - Individuals remain as passive parameters in the workings of institutional systems and the exchanges between them.
  - Individuals have no independent existence – they remain scattered and fragmented across institutions, QHINs/HINs, systems and devices.

- The need is to have an individual's health information as a coherent entity we access/use and contribute to – but that is non-existent!
- There is no basis for informed and coordinated individual-centered health care or self-care.

Misalignment

- The obsession with TEFCA distracts from the underlying architectural misalignment between information processes and individual health care itself.
- The answer to tackling the complexity is not more of the same. The workings of institutions cannot effectively act as proxies for the experiences of an individual. The perspective on the problem needs reorienting:
  - There is no need for institutions to interoperate directly with each other for purposes beyond immediate business relationships.
  - There is a need to provide coordinated and embracing care to all patients that actively and continuously serves their individual health and healthcare needs.
- The conceptual misalignment of TEFCA leads to two categorical errors:
  1. Trying to solve a problem that does not need solving and is effectively *intractable*
     - As evidenced by the TEFCA Scatter Model; and
  2. Failing to solve the one that does need solving and is *tractable*
     - Establishing the individual at the center.

- The breakthrough will come once we embrace the centered model and put the scattered model to rest.

Tractable vs. Intractable, Centered vs. Scattered

	Centered = HRB/PHR Model	**Scattered = TEFCA Model**
Basics	Single designated and secure system which is: • Patient-controlled and provider neutral • Maintained by a trusted custodian organization Where the patient or their representative: • Maintains unique ID, account and address • Designates a single place to send/store their records, e.g., after each encounter	Patient data/records are managed across 10s and 100s of HINs and 1000s of systems/ devices, each of which maintains/manages: • Trusted software and storage • Accountability, authentication, authorization, consents, access control, audit mechanisms • Some fragment of the patient record
Broadcast query	Query is directed to a single designated HRB custodian and account for each patient	Query goes to 10s or 100s of HINs, then on to 1000s, 10,000s, 100,000s of systems/devices
Query response	• Response is immediate • All relevant and permitted records are immediately available • You immediately know what you need to know	• Response may be nothing, trickle or deluge • Response content may vary with each query • Response may take minutes, hours or days • You don't know what you don't know • You don't know how long to wait
Confidentiality/ Authorization	Managed at a single point by each patient and/or patient representative/coach	Managed within a complex lattice of provider and HIN permissions plus patient consents
Real-time + Continuous	Sustained (24 x 7) support for individual health and healthcare – monitoring and guidance	Never, never land

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A Center for the Individual – Making It Real

- No amount of information standards, policy frameworks or common agreements will answer these "Scatter Model" conundrums.

- The problem that needs solving is making the “center” real: by creating an HIT infrastructure that “primarily promotes and serves the individual”.
- Providers then interact with that system as a routine part of care and as part of the duty of care, contributing to a persistent coherent record of an individual's overall health and healthcare, their journey and path forward.
- Providers make use of that record when needed – without recourse to interactions with other institutions.
- Providers continue to operate their own EHR/HIT systems for their own purposes – established as their business record – capturing the operational details of the care they provide and activities they perform.

A Center for the Individual – Joining the Carer and the Cared For

- The aim is to ensure individuals get coherent, “joined-up” care.
- This can only be achieved if the individual is the conceptual design center of our HIT infrastructure.
- Care is provided to individuals and hence information architecture should align with that care paradigm.
- Exchanges should be between the individual and those providing them with care.

- Providers should stop talking *about* individuals and instead focus on talking *with* them.

A Center for the Individual – Many to Many <becomes> Many to One

- Pursuit of TEFCA and the “Scatter Model” will continue to distract from the
  - Fundamental need to architect a new individual-centered HIT infrastructure that is distinct from, but works with, institutional systems.
- With the proper architecture, the
  - Many-to-many problems created by myriad QHINs/HINs/HIEs can be transformed into a
  - Many-to-one architecture with the one being the individual at the center.

A Center for the Individual – Need: Continuous Monitoring/Guidance

- Individual health and care
  - Should be (able to be) supported continuously – without interruption or gaps
  - Enabled by real-time monitoring and prompts (guidance)
  - Even when ‘no one is looking’ = independent of institutional awakenings

- Currently there is no provision for – and no system available and working 24 x 7 – in the space between providers – where individuals spend their entire lives

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“For TEFCA to fulfill its potential, it must be built on a foundation that prioritizes the individual. Ensuring transparency and meaningful control isn't just a technical feature; it's a fundamental requirement for an ethical and effective national health data exchange that truly serves the people it's meant to benefit.” – Josh Mandel MD, via LinkedIn

Agree with Dr Mandel that “transparency and meaningful control” for each individual is a key objective, yet disagree that TEFCA is the right fulfillment vehicle.

A Center for the Individual – Effectiveness of the Centered Model

	Centered	**Scattered**
Information Practices, Governance		
Rules and Logs • User authentication • Authorization, consents for information sharing • Access control • Minimum necessary and "need to know" • Audit events • Data exchange events	• Consistent, uniformly implemented • Can be controlled by patient, patient representative/coach Always: • Visible to patient • Visible to clinicians, as permitted	Across multiple source systems/devices, HINs/QHINs: • Disparate conventions, policies, governance • Controlled by multiple entities • Unevenly implemented Rules are: • Mostly invisible to patient • Mostly invisible to clinicians
Personal Parameters		
Personal • IDs, identity parameters • Demographics • Genetics • Social determinants of health • Insurance, coverage • Next of kin • Advance directives, health-related consents	• Consistent, uniformly implemented • Visible and can be managed by patient, patient representative/ coach	Across multiple source systems/devices, HINs/QHINs: • Disparate conventions, policies, governance • Controlled by multiple entities • Unevenly implemented
Clinical Findings		
Current and Historical • Encounters • Personal and family health history & physical findings • Vital signs • Assessments • Progress notes • Care records • Diagnostic results	Patient data in one place • Can be monitored directly by the patient • Can be annotated or corrected by the patient	• Patient data in multiple disparate fragments across multiple source systems/ devices, HINs/QHINs • Multiple portals and/or repeated "pulls" to gather "current" view
Coordinated Care		

Current and Historical
• Care Plan(s)
• Orders
• Therapies
• Consults
• Referrals
• Schedules
  • Appointments + followup
• Messages, interactions with caregivers
	Uniform, integrated
	• All patient data
	• All providers
	• All activities
	  • Past, present, future
	In real-time, patient can:
	• Monitor data and actions
	• Be prompted to take actions
	• Complete and document actions taken
		• Patient data in multiple disparate fragments across multiple source systems/ devices, HINs/QHINs
		• Multiple portals and/or repeated "pulls" to gather "current" view

Reconciled Lists		
Current and Historical • Medications • Problems, conditions • Allergies, intolerances • Immunizations • Immunization forecast • Procedures and interventions	Patient data in one place In real-time, patient can: • Monitor data and actions • Curate lists • Be prompted to take actions • Complete and document actions taken	• Patient data in multiple disparate fragments across multiple source systems/ devices, HINs/QHINs • Multiple portals and/or repeated "pulls" to gather "current" view

[The preceding table, in part, corresponds to the response to PC-1b and PC-2b above]

Individual PHR or HRB Account – Account and Data Exchange Functions

Account Setup
1. Patient or patient representative/coach establishes an individual account with trusted entity, such as an independent Health Record Bank (HRB) or Personal Health Record (PHR) Entity a. Patient receives: i. A Unique ID ii. An Account Access URL to their individual account (with modifiable password) iii. A FHIR Endpoint (Push URL) where their health record entries can be sent (in batches conforming with HL7 FHIR) 2. Patient shares the Push URL with their healthcare Providers
Account Deposits
3. Account "deposits" are made as persistent health record entries: a. Originated by patients, their guardians or representatives b. Originated by, then received from, external sources, such as patient healthcare providers (e.g., at the conclusion of a patient encounter) 4. All "deposits" are visible to the individual patient
Account Withdrawals
5. Account "withdrawals" are copied from persistent health record entries: a. Based on inquiries received from/transmitted to: i. Covered entities legitimately operating under HIPAA provisions for Treatment, Payment and Operations ii. Other entities authorized by the patient or their representative • Patient Representatives, Health Coaches, Family Members, Friends... • Clinical Trials/Researchers, Public Health, Marketing... b. Based on "push" or "pull" model c. Extracts may be constrained, as appropriate, to: i. "Minimum necessary and "need to know" for intended purpose of access/use ii. De-identified data 6. All "withdrawals" are visible to the individual patient

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Record Entry Convention

- From the HL7 FHIR Record Lifecycle Event Implementation Guide:
  - Actions are taken to support patient health.
    - Actions are taken in provision of healthcare to individuals.
    - Actions are taken as the result of rules-based EHR[/HIT] System algorithms.
    - Actors (i.e., patients, providers, users, systems) take Actions.
    - Actions broadly encompass tasks, acts, procedures or services performed or provided.
  - The EHR[/HIT] System captures Actions taken and creates corresponding Record Entries.
    - Record Entries provide persistent evidence of Action occurrence, context, disposition, facts, findings and observations.
    - From the point of Record Entry origination to the end of its lifespan, the EHR[/HIT] System manages each Entry consistent with and according to scope of practice, organizational policy, and jurisdictional law.
    - In support of individual health and in provision of healthcare to individuals, Actors perform Actions and Actions have corresponding Entries in the EHR[/HIT] Record, (i.e., Action instances are documented by Record Entry instances).
    - Record Entries may be captured during the course of the Action or sometime thereafter.
    - The Actor (author/source) of the Record Entry may be the same as an Actor performing the Action or not…

- Actions have associated metadata (e.g., who, what, when, where, why, how, under what conditions, in what context).
  - The corresponding Record Entry captures this metadata along with other Action and Record Entry related information.
- Each Record Entry also includes its own provenance metadata such as who (authoring Actor) and when (documented).
  - Record Entries may be encapsulated to bind Actor (individual, organization, and/or system) signatures to data and metadata content and data/time of occurrence.
  - Actions and related Record Entries capture a chronology of patient health and healthcare and also a chronology of operations and services provided in/by a healthcare enterprise.
  - Record Entries reflect changes in health information from the time it was created, to the time it was amended, sent, received, etc. In this manner, each Record Entry serves as persistent evidence of an Action taken, enabling providers to maintain comprehensive information that may be needed for legal, [clinical,] business, and disclosure purposes.
  - To satisfy these purposes, Record Entries must also be retained and persisted without alteration.
- Record Entries have both a lifecycle and a lifespan.
  - Lifecycle Events include originate, retain, amend, verify, attest, access/view, de-identify, transmit/receive, and more.
  - Lifecycle Events occur at various points in a Record Entry lifespan, always starting with a point of origination and retention (i.e., when the Entry is first created and stored).
- A Record Entry may have a pre and post Event state if content is modified.
  - In this case, the original Record Entry is preserved (with signature binding) and a new Entry is created (with new signature binding).
  - A Record Entry contains data and metadata, in multiple formats, following various conventions and standards. Included data may be tagged, and/or delimited, structured (concise, encoded, computable), or unstructured (free form, non-computable).
  - Data may be encoded as text, document, images, audio, waveforms, in ASCII, binary or other encoding.
  - [Record Entry content may encompass, correspond or be bound directly to FHIR resource(s) related to Actions Taken.]

- To effectively manage health data/records longitudinally and continuously throughout the continuum of care, needed is the Record Entry convention implemented across EHR/HIT systems and points of exchange. These conventions are fully specified in the standards enumerated in response to PC-12 below.

Globally Unique ID (GUID) for Record Entries

- Needed is a GUID encapsulated with Record Entries, whether originated in EHR System (provider facing), originated in PHR/HRB System (patient facing) or originated elsewhere
  - GUID-encapsulated Record Entries are immutable
  - Record Entries may be shared but the GUID itself is unique and universal
  - Record Entries may be amended/updated/annotated against original GUID, as a new Entry encapsulated as an extension of that GUID

Note that the English National Health Service is exploring feasibilities/capabilities for a single patient record (SPR) as solicited in this RFI, issued 5 May 2025.

PC-10a – Please provide specific examples?

[See responses to PC-10 above]

PC-10b – What changes would you suggest?

[See responses to PC-10 above]

- TEFCA is a vestige of the big government/big institution/big vendor complex, built to serve their interests, not the interests of the individual patient, clinician or caregiver, also not small/medium and rural providers.
  - Suggest to transition our foundational HIT infrastructure from scattered model (TEFCA) to centered model: one patient one record, single source of truth for each patient.
  - Suggest to set aside current TEFCA and its scattered (fragmented, dissonant) architecture.
- Why are hospitals and providers everywhere closing, going bankrupt, with many others in financial despair? Unfathomable mandates and intrusions on their businesses which require major $$$ outlay but little value, such as TEFCA, play a major role.

PC-10c – What use cases could have a significant impact if implemented through TEFCA?

[See responses to PC-10 and PC-10a/b above]

PC-10d – What standards are you aware of that are currently working well to advance access and existing exchange purposes?

- FHIR APIs – <u>sans TEFCA</u>.

PC-10e – What standards are you aware of that are not currently in wide use, but could improve data access and integration?

[See standards listed in the response to PC-12c below]

PC-10f – Are there redundant standards, protocols, or channels that should be consolidated?

- Use current generation standards – HL7 FHIR w/APIs – to consolidate and supplant prior generation standards: e.g., HL7 Clinical Document Architecture (CDA), Consolidated CDA and HL7 v2 messages.

PC-10g – Are there adequate alternatives outside of TEFCA for achieving widespread patient access to their health information?

- TEFCA <u>does not</u> currently achieve "widespread patient access to their health information".
- Promote individual PHR and Health Data Record accounts – one patient one record, single source of truth.
- Set aside complex many-to-many exchange paths (scattered model of TEFCA) in favor of many-to-one (patient-centric PHR/HDB accounts).

PC-11 – How are health information exchanges (HIEs) currently helping to advance patient access to health information in the real world?

- Most HINs implement previous generation data exchange standards (based on HL7 v2 messages, HL7 CDA/CCDA documents) and are thus constrained by their limitations.

PC-11a – How valuable, available, and accurate do you find the data they share to be?

- So the question is whether shared data has value, is available and is accurate. Value is an equation resolving to whether "the data" has appropriate qualities to be fit for its intended purpose/use, noting that the same data may be fit for one purpose/use but not another.
- Data qualities must be assured and verifiable in all cases. In the real world, shared data often comprises a dump of elements which defy ready evaluation of vital data qualities due to a lack of key metadata, attribution, context, attestation (e.g., encapsulated content bound to author's and/or verifier's signature), or other reliable evidence of integrity and trustworthiness.

Following is a compilation of key characteristics, as prepared by the HL7 EHR Work Group, Data Quality Project Team...

- Data Quality is Vital to Data Usability
  - What are the vital qualities (characteristics) of data that promote usability (use, usefulness)?
  - Is the data true (factual, accurate, authentic) and trustworthy (reliable, assured)?

- Is the data identifiable (known subject and provider of care)?
- Is the data attributable (known source and authorship)?
- Is the data timely (timestamped, current)?
- Is the data traceable (to known source, via trusted pathway and conveyance)?
- Is the data immutable (indelible, persisting without substantive alteration)?
- Is the data in context (content bound to context)?
- Is the data relevant (pertinent, applicable to end use)?
- Is the data digestible (concise, comprehend-able)?
- Is the data fit for purpose (purpose of capture compatible with purpose of use)?
- Is the data ready and action-able in support of real-time care delivery and/or a specific end use?

- To avoid concerns regarding burden, let's presume that vital data qualities are present and managed by underlying system and data exchange infrastructure but not necessarily visible in the initial "top view" – i.e., evidence of these qualities might be accessed via a "drill-down view" capability.

Vital Data Qualities	**Capabilities and safeguards must ensure health data/records...**
***Known* and verified (verifiable) as to identity**: • Subject: patient • Provider: individual and organization • Systems, devices and software	Are associated with the correct identity and subject (of care/treatment)
***Known* and tied to actions taken: who did what, when, where and why** • Actions are taken to support individual health, to provide healthcare... • Actions are taken by an accountable agent (person, organization, system (software/device)), at a date/time, for a period of time, at a location (physical, network, device), for a purpose • Data resulting from actions taken includes facts, findings, observations...	Show a clear relationship/binding of datasets/data elements with actions taken – who took what action, when, where and why
***Known* to be oriented in time: date/time of occurrence, chronology, sequence** • What has happened: past, retrospective • What is now in progress: present, concurrent • What is anticipated, planned: future, prospective	Show time orientation and chronology/ sequence
***Known* to retain clinical context and maintain vital inter-relationships** among/between (as applicable): • Problems, diagnoses, complaints, symptoms, encounters, history and physical findings, allergies, medications, vaccinations, assessments, goals/objectives, clinical decisions, orders, results, diagnostic procedures, interventions, observations, treatments/therapies, referrals, consults, outcomes, protocols, care plans and status...	Show clear inter-relationships/ binding between datasets/data elements and their clinical context (as noted in the left-hand column)
***Known* as to source and provenance ("source of truth")**, with traceability to point of origination: human, device, software	Show dataset/data element provenance with traceability to source/point of origination
***Known* as to accountable human authorship** (if applicable) with role and credentials	Show dataset/data element authorship with role and credentials, as applicable
***Known* to be verified/attested (or not)** with evidence of verification/attestation, verifier(s)/ attester(s), date(s)/time(s) and method(s), for example: • Human verification of data sourced by automated device • Preceptor verification of data sourced by student • Attestation of facts/findings to substantiate claim for payment	Show evidence of dataset/element verification, as applicable

Vital Data Qualities	**Capabilities and safeguards must ensure health data/records...**
***Known* to be updated (or not)** with evidence of prior state(s), effective date(s)/time(s) • Non-destructive update(s) over time	Show evidence of dataset/data element update, prior state(s), effective date(s)/time(s), as applicable
***Known* to be unaltered** (maintaining fidelity to original/source content) OR ***Known* to be altered/transformed** from source content/ representation to another • From one coding/classification system to another, or • From one human language to another	Show evidence of dataset/data element non-alteration or alteration, as applicable
***Known*, if altered/transformed, to carry original content/representation alongside (or not)**	Carry original dataset/data element content (or not), as applicable
***Known* to be complete** or ***Known* to be incomplete, partial or pending** or ***Known* to be a snippet/fragment** with other essential details elsewhere or ***Known* to be a header, pointer or link** to essential content elsewhere	Show notation of dataset/data element completeness (or not), as applicable
***Known* to be associable and congruent with like data**, having same/similar context • Comparable, correlate-able, trend-able	Have the same/similar context so as to be comparable, even/especially if sourced by separate EHR/HIT systems
***Known* to be consistent and uniform – in terms of data definition – with corresponding data**: • Common element name(s), common data type(s), common range (including normal or reference range), common input/display/storage format, common unit(s) & scale of measure, common vocabulary, common codes & value sets	Have consistent data naming and definition, even/especially if sourced by separate EHR/HIT systems
***Known* to be sourced as structured (coded) content or not**	Show data source as structured content or not
***Known*, if coded, to include:** • Coding convention – vocabulary/terminology set or value set – and version	Show coding convention and version, if applicable
***Known* as to method and purpose of data capture**	Show method and purpose of capture, if applicable
***Known* as to intended purpose of data use**	Show intended purpose of use, if applicable
***Known* if captured from external sources**	Show external data source, if applicable
***Known* as to how external data is integrated (incorporated) with OR maintained separately from** health data/records in local EHR/HIT system	Show data integration or segregation, as applicable

- With some exceptions in today's HIT infrastructure, the capture and conveyance of vital data qualities are an after-thought, if thought is given at all.

- Instead such evidence must be pre-planned and integral to all health data/record flows, from data source (point of origination) to end use (each ultimate point of data access/use), including all points of exchange in between.

PC-11b – What changes would you suggest?

- Suggest the full set of data qualities be promoted as requisite in all solutions and become evident in all aspects of health data/record management from source to use, as per the table in PC-11a above.
- This evidence is mostly about what is captured at the data source (point of origination) and conveyed forward from there. Once the moment of capture has passed, such evidence is mostly lost and cannot recovered without extraordinary effort, if even then possible.

Fax or Faux? Are We Really Ready to Replace the Fax in Health Information Sharing?

Fax → a facsimile; an exact copy, especially of a document; an image of a document made by electronic scanning and transmitted as data by telecommunication links

Faux → artificial; not genuine or real; being an imitation of the genuine article

While there is widespread agreement with the objective to replace fax machine transmission/reproduction of health records with "interoperability" between EHR/HIT systems, the question is whether the current set of interoperability schemes are in fact competent to achieve/replace a basic fax function. Indeed there is one vital characteristic where it remains superior, if not infallible... What goes in (source artifact) is identical to what comes out (artifact produced by the fax). If not, these devices would be immediately placed out of service and then repaired or replaced.

The design of today's interoperability schemes has largely disregarded the requirement to reproduce (for purposes of exchange and end use) an identical copy of the original (source) record, as an authentic reproduction without alteration.

Today's interoperability schemes are mostly comprised of "standardized" exchange artifacts, e.g., HL7 v2/v3 messages, CDA/C-CDA documents or FHIR resources. Use of these artifacts typically REQUIRES data content transformation twice in the course of exchange: first at transmission, from source representation to exchange artifact representation, then again on receipt, from the exchange artifact to receiver representation. Thus: Source → Exchange → Receiver

Transformations often INTRODUCE alterations, omissions and unresolved errors in data content. Transformations often LOSE OR DISJOIN attribution, authorship/signature, verification/attestation, provenance, context, purpose of capture, consistency, comparability, units of measure and binding to conventions of coding, classification and vocabulary systems.

The risks (to data integrity) introduced by data transformation – as described above – also represent major risks to the integrity of the clinical process and most importantly, SERIOUS RISKS TO PATIENT SAFETY.

These risks also strike at core principles of trust (assurance) and truth (authenticity) which are essential to achieve full interoperability and usability of, and confidence in, health information – for providers and individual subjects of care alike.

Suggest that it's time to fully assess and address serious risks to clinical integrity and patient safety posed by health data/record management and exchange "solutions" that fail to maintain fidelity to source.

Truth and Trust – Measuring Interoperability

- So how do we measure the achievement of interoperability?
  - In terms of fitness for Primary Use? Secondary Use?
  - When viewed at each ultimate (downstream) point of health data/record access/use?
- With achievement based on:
  - Truth = authentic, factual
  - Trust = assurance, certainty, reliance
- Evidence of Truth (at each ultimate point of access/use) is the key Trust determinant.

Source of Truth for Health Data/Record Content

- Truth for all primary use – clinical care, interventions and decision making – is unaltered source health data/record content.
- The receiving clinician will first and always trust (rely on) this direct evidence of clinical facts, findings and observations.
- Data integrity (including fidelity to source) is fundamental to all aspects of clinical integrity and patient safety.

The Challenge of Data Transformation

- True interoperability ensures fitness for purpose/use at each ultimate point of health data/record end use. The following table illustrates the challenge of health data/record exchange between heterogeneous systems and the risk to fitness (for purpose/use) posed by data transformations. Double transformations often occur during the course of exchange when health data/record content is transformed to/from exchange artifacts (e.g., HL7 v2 messages, HL7 CDA/CCDA documents or HL7 FHIR resources) – once by the source/sending system and once again by the receiving system.

Use	Typical Scope	Health Data/Record Exchange			Post Exchange Fit for Use?
		Source	→ → →	Receiver	
Primary	• Care, care coordination • Treatment • Diagnosis • Decision making • Interventions ...	Without Transformation (maintaining/ensuring fidelity to source)			**YES**
		With Transformation(s)			**Often NO**
Secondary	• Billing: claim, payment, substantiation • Operations • Public Health • Research • Legal Proceedings • Oversight, Accreditation ...	With Transformation(s)			**Typically YES**

- The consequences of data transformation are mostly ignored by current HIE, source and receiver exchange implementations – with or without TEFCA – yet transformation(s) engage an opportunity for the *introduction of alterations, errors, omissions and disjunctions in health data/record content*. Such are a risk to clinical integrity and more importantly to patient safety.

Transformation during Exchange…	typically Alters…	and may INTRODUCE Errors, Omissions and Disjunctions in…	Health Data/Record Content

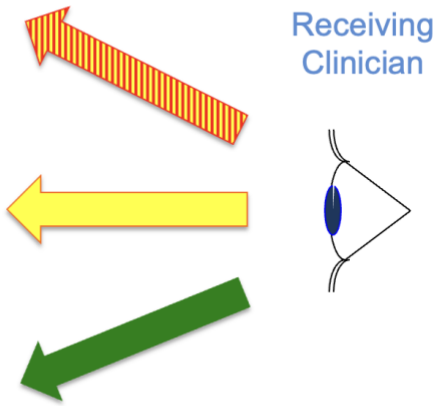
Consider the following...

Potential Alteration, Errors, Omission and Disjunctions

Examples	Source Clinical Content is/has…	Likely Disjunction	
Mismatched	Incorrectly matched • Including Patient or Provider identity	Error	
	Structured content mapped to/from unstructured content	Error or	Alteration
	Disjoint data types: e.g., integer vs. decimal	Error or	Alteration
	Codes/values mapped one to many	Error or	Alteration
Incomplete or missing	No corresponding target data element	Omission	
	No corresponding code/value in target code/value set	Omission or	Alteration
Less Precise	Source codes/values mapped many to one	Error or	Alteration
	Less digits/characters, rounding/truncation	Error or	Alteration
Skewed	As the effect of multiple transforms • 1 off + 1 off + 1 off + 1 off...	Error or	Alteration

The Clinician View (post transformation) – What is truth? How am I assured?

Transforms	Primary Use – Clinician View
1, 2, 3, 4…	1 2 3 4 Blind Transforms View Last (Sum) Result Use with Extreme Caution!
	1 2 3 4 Visible Transforms View each Result Be Aware!
0	View Unaltered Source Health Record Content Be Assured!



- Suggest that whenever data transformation occurs, it be logged to allow continuous monitoring and corrective updates to software and mapping algorithms, where possible.

Foundations of Trusted Exchange (Interoperability)

- Truth = factual, authentic
  → Facts are evident
- Trust = trustworthiness = assurance, reliance
  → I am assured, I trust, I rely on (in the first person)

- To achieve interoperability, truth and trust are essential
  As evidenced at each downstream point of access/view – to each primary or secondary user of health data/records
  → Primary use = Clinical care, interventions, decision making...
  → Secondary use = All other purposes

Truth	**as evidence for**	**Trust**
• Identity is evident		
• Actions are evident: e.g., actions taken to support individual health and provide healthcare		
• Who took what action when, where and why is evident		
• Action facts, findings and observations are evident		Establishing: • Safety • Belief (believability) • As a conscious human conclusion (conviction) • Based on – and manifest in – evidence presented • Traceable to the "source of truth" Resolving to: • Certainty: sureness • Reliance: placing trust in
• Source, origination and provenance is evident		
• Verification and/or attestation (signature) is evident (confirming accuracy/completeness)		
• Signature/content binding is evident		
• Who authored what when, where and why is evident		
• Content is un-altered		
• Context is evident		
• Completeness (or not) is evident		
• Update(s) to original content are evident		
• Chain of Trust is evident • From source to use		
• Transformation(s) are evident: e.g., to/from exchange artifacts		
• Original "Source of Truth" is evident		

End-to-End Chain of Trust

- To ensure data quality and usability, data must be conveyed via a verifiable chain of trust from **source to end use**:
  - Starting as captured at the **source** (point of origination), then
  - Retained in the source EHR/HIT system, then
  - Transmitted from the source system, then
  - If so configured: Received, retained and ultimately transmitted from intermediary system (e.g., HIN), then
  - Received and retained by the receiving EHR/HIT system, then
  - Made available to each ultimate **end use and user** (point of access/use).

End-to-End Data Flow example...

	Health Data/Record Chain of Trust from Point of Collection to each ultimate Point of Use to Support the <u>Affirmative Trust Decision for Primary Clinical Use</u>					
	Flow	Record Lifecycle – Point of Health Data/Record…	(For primary clinical use)	Audit Event	Provenance Event	Original Content
	Source System					
COLLECT	↓	Collection/capture – Point of Origination • Source of Truth • Anchor Point for Chain of Trust	• Clinical facts, findings and observations are captured • Clinical context is captured • Provenance is captured: • Who, what, when, where, why • Identities are established: • Patient: subject of care • Provider: organization, individual • Author of data/record content	**Yes**	**Yes**	Is captured
	↓	Retention	Of Source Record Entry	**Yes**	No	Is retained
	↓	Attestation	• Application of Signature • Bound to data/record content	**Yes**	**Yes**	Is attested (and signed)
SHARE	↓	Transformation	From Source Record Entry to Exchange Artifact (e.g., HL7 message or document or FHIR resource)	**Yes**	**Yes**	Is carried
	↓	Transmission	Of Exchange Artifact	**Yes**	No	Is carried
	Receiving System					
	↓	Receipt	Of Exchange Artifact	**Yes**	No	Is carried
	↓	Transformation	From Exchange Artifact to Receiver Record Entry	**Yes**	**Yes**	Is carried
	↓	Retention	Of Receiver Record Entry	**Yes**	No	Is retained
USE	↓	Access, view • Trust Decision	By End User	**Yes**	No	Is accessed, viewed

- Suggest establishing common unit of record, as described in in the Record Entry Convention in response to PC-10 above.
- Suggest formalizing health data/record flow to ensure content and context integrity end to end, source to end use.

[See response to PC-12 below, including standards referenced]

Chain of Trust – The Challenge of Testing, Verification, Certification – Within + Across EHR/HIT Systems

Chain of Trust – EHR/HIT System Testing, Verification and Certification Program		Truth → → Chain of Trust → → Trust									
		Source EHR/HIT System					→	Receiving EHR/HIT System			
At Point of Health Data/Record >		Capture, Origination (Source of Truth)	Retention	Attestation, apply/bind signature, to content	Transformation to Exchange Artifact	Transmit Exchange Artifact, including original	Exchange	Receipt of Exchange Artifact	Transformation from Exchange Artifact	Retention	User Access/View (Trust Decision)
Test, Verify, Certify	Source Role	Systems A & B Separately									
	Receiver Role							Systems A & B Separately			
	System A → B	System A					→	System B			
Implement A → B		System A					→	System B			
Live A → B		System A					→	System B			
Ensuring Chain of Trust A → B		Original Content → → → → → → →									

- Suggest expanding scope of testing, verification and certification to both within and across EHR/HIT systems.

End-to-End Traceability

- Tied with the chain of trust and lacking in our current HIT infrastructure is true traceability:

Downstream	• From source perspective – as viewed by clinician, author or patient • From point of origination/authorship downstream • To whence my health data/record content goes • Across all points of exchange	→→→
Upstream	• From receiver perspective – as viewed by end user, clinician or patient • From point of access/view upstream • From whence this health data/record content comes • Across all points of exchange	←←←

- Suggest formalizing data flow and methods to ensure bidirectional traceability.

Who did What When, Where and Why

The most basic of EHR System functions is to manage the EHR Record itself: as a persistent chronicle (evidence) of Actions taken in support of individual health and provision of healthcare. These Actions are typically documented as EHR record entries, readily indexed and forming a chronology of:

	Who	(did) What	When	Where	Why
1	Action taken...				
	Actor(s)	Took Action	At Action date/time (with duration)	At Action Location	To fulfill Action Reason or Purpose
2	Then, Action documented in EHR/HIT record entry...				
	Actor(s) as Author, Enterer	Recorded Action Details, including Facts, Findings, Observations	At Recording date/time	At Recording Location	To create EHR/HIT record entry documenting Action taken

- Actors are accountable for Actions taken.
- Actors are accountable to document (provide evidence of) those Actions by capturing authentic facts, findings and observations in EHR/HIT record entries.
- Actor(s) taking the action may be the same or different than the author(s) of the health record entry (recording the Action taken): e.g., a clinician may do both, alternatively a clinician may take the Action and a scribe may record it.
- Action is taken and recorded simultaneously or the Action is taken and recorded sometime later.

Health Record Trustworthiness - as applicable to Actors, Roles and Accountabilities

- As we consider trustworthiness, it is crucial to assess accountabilities and understand the relationship between actors, their role and participation, and the target of their efforts in terms of actions taken to support health and provide healthcare, and interactions with health data/records and EHR/HIT systems. See following example – which is intended to be representative not exhaustive...

Actor – Who	Role/Participation	Target	Frequency – When
Individuals	As subject of... As performer of... As observer/witness of...	Action taken	Per occurrence
	As subject of... As author/originator of... As scribe/enterer of... As verifier of... As attester of...	EHR/HIT record entry content	
Software – systems, devices	As source of...		
Organizations	As performer of...	Action taken	
	As steward of...	Persistent EHR/HIT system-based clinical/business record	Continuously, uninterrupted
→ Audit/Legal	As inquirers, reviewers of...		Periodically
→ Providers	As implementer of...	EHR/HIT system software	At setup
	As configurator of...	EHR/HIT system parameters	Periodically
→ Vendors	As developer/proprietor of...	EHR /HIT system software	Ongoing

- Suggest formalizing a trustworthiness framework around accountability for actions taken to support health and provide healthcare, for interactions with health data/records and with EHR/HIT systems. Accountable interactions include originate, update, verify/attest, access/view health data/record content and more.

Lifecycle Events in Health Record Use Case Scenarios - Both Manual (Paper) and Electronic

- Another key area of concern is the (almost confounding) proliferation of both physical and digital artifacts that are inherent in our management of electronic health data/records, particularly when considered in contrast to traditional manual record-keeping.

- Let's follow the progression of health data/records - both manual (including paper) and electronic - in several scenarios from point of origination (collection, capture) to the point of access/use.

- Note that the culminating lifecycle event (far right in these example scenarios) is the point of Access/View, where a trust decision must be made regarding whether the health data/record content received is of sufficient integrity to be the basis for trust, decision-making and actions to be taken.

- In the following scenarios, progression is left → right.

Lifecycle Events in a Basic Manual Health Record Use Case – Within One Domain

	(Real World)	Collect			Use
Lifecycle Event >	N/A	Originate	Verify/Attest	Retain	Access/View (Trust Decision)
Flow >	→	→	→	→	
Scenario 1A - Manual Record - One Domain					
Actor >	Human	Human	Human	Human	Human
Manual Record >	Takes action(s) to support individual health and deliver healthcare: observes, measures, assesses	Writes Entry in Source Paper Record	Signs Entry in Source Paper Record	Saves/Files Source Paper Record	Views Copy
Artifact >		#1 Original	#1 Original	#1 Original	#1 Original

Lifecycle Events in a Basic Electronic Health Record Use Case – Within One Domain

	(Real World)	Collect			Use
Lifecycle Event >		Originate	Verify/Attest	Retain	Access/View (Trust Decision)
Flow >	→	→	→	→	
Scenario 1A - Manual Record - One Domain					
Actor >	Human	Human	Human	Human	Human
Manual Record >	Takes action(s), observes, measures, assesses	Writes Entry in Source Paper Record	Signs Entry in Source Paper Record	Saves/Files Source Paper Record	Views Copy
Artifact >		#1 Original	#1 Original	#1 Original	#1 Original
Scenario 1B - Electronic Record - One Domain					
System >		EHR-A			
Actor >	Human	Human	Human	System	Human
Electronic Record >	Takes action(s) to support individual health and deliver healthcare: observes, measures, assesses	Documents action(s), observations, measures, assessments	Verifies action(s)/signs observations, measures, assessments	Saves EHR-A Record Entry	Views action(s) taken, observations, measures, assessments
Artifact >		#1 Data entry screen/window	#2 Verification screen/window	#3 EHR-A Record Entry	#4 Presentation screen/window
State >		Is Transient	Is Transient	Is Persistent	Is Transient

[Additional scenarios follow...]

Lifecycle Events in a Basic Electronic Record Print/Output Use Case - Within One Domain

	(Real World)	Collect			Share	Use
Lifecycle Event >	N/A	Originate	Verify/Attest	Retain	Print/Output	Access/View (Trust Decision)
Flow >	→	→	→	→	→	
Scenario 2A - Manual Record - One Domain - w/Print						
Actor >	Human	Human	Human	Human	Human	Human
Manual Record >	Takes action(s), observes, measures, assesses	Writes Entry in Source Paper Record	Signs Entry in Source Paper Record	Saves/Files Source Paper Record	Duplicates Source Paper Record via Photocopy	Views Copy
Artifact >		#1 Original	#1 Original	#1 Original	#2 Copy of Original	#2 Copy of Original
Scenario 2B - Electronic Record - One Domain - w/Print						
System >		EHR-A				
Actor >	Human	Human	Human	System	System	Human
Electronic Record >	Takes action(s) to support individual health and deliver healthcare: observes, measures, assesses	Documents action(s), observations, measures, assessments	Verifies action(s)/signs observations, measures, assessments	Saves EHR-A Record Entry	Renders Hard/Soft Copy	Views action(s) taken, observations, measures, assessments
Artifact >		#1 Data entry screen/window	#2 Verification screen/window	#3 EHR-A Record Entry	#4 Hard or Soft Copy	#5 Presentation screen/window
State >		Is Transient	Is Transient	Is Persistent	Is Transient?	Is Transient

[Additional scenario follows...]

Lifecycle Events in a Basic Electronic Record Exchange Use Case - Across Two Domains

	(Real World)	Collect			Share			Use
Lifecycle Event >	N/A	Originate	Verify/Attest	Retain	Transmit	Receive	Retain	Access/View (Trust Decision)
Flow >	→	→	→	→	→	→	→	
Scenario 3A - Manual Record - 1+ Domains								
Actor >	Human	Human	Human	Human	Human	Human	Human	Human
Manual Record >	Takes action(s), observes, measures, assesses	Writes Entry in Source Paper Record	Signs Entry in Source Paper Record	Saves/Files Source Paper Record	Transmits Copy of Source Paper Record via Fax	Receives Fax or Photocopy	Saves/Files Fax or Photocopy	Views Copy
Artifact >		#1 Original	#1 Original	#1 Original	#2 Copy of Original	#2 Copy of Original	#2 Copy of Original	#2 Copy of Original
Scenario 3B - Electronic Record - 1+ Domains								
System >		EHR-A				EHR-B		
Actor >	Human	Human	Human	System	System	System	System	Human
Electronic Record >	Takes action(s) to support individual health and deliver healthcare: observes, measures, assesses	Documents action(s), observations, measures, assessments	Verifies action(s)/signs observations, measures, assessments	Saves EHR-A Record Entry	Sends Exchange Artifact	Receives Exchange Artifact	Saves EHR-B Record Entry	Views action(s) taken, observations, measures, assessments
Artifact >		#1 Data entry screen/window	#2 Verification screen/window	#3 EHR-A Record Entry	#4 Exchange Artifact (message, document, resource)		#5 EHR-B Record Entry	#6 Presentation screen/window
State >		Is Transient	Is Transient	Is Persistent	Is Transient?		Is Persistent	Is Transient

- Note that clinical activities, including care, treatment, decision-making and documentation of facts, findings and observations typically occur at the first (originate) and last (access/view) lifecycle events, yet those artifacts are transient (not persistent) in our representative electronic health data/record flow, based on today's HIT infrastructure

Suggest formalizing a framework for health data/record flows and artifacts produced (rendered) at each lifecycle event, including those persistent and transient. To ensure integrity and trustworthiness, it is crucial that standards development and adoption be based on this framework.

Health Informatics Standards Board (HISB)

Suggest creation of a new Health Informatics Standards Board (HISB), as articulated by Joe Schneider MD:

- "Replace government-directed health informatics requirements with a private-sector governing body (the 'Health Informatics Standards Board' - HISB):

- "HISB would be the sole authority in promulgating health informatics standards developed by existing standards bodies that must be used in all health-related electronic systems, including apps, EMRs, and HIEs;
- "HISB's relationship with ASTP/ONC would be analogous to how the Securities and Exchange Commission defers to the Financial Accounting Standards Board (FASB) for the development and promulgation of Generally Accepted Accounting Principles (GAAP) and serves only as the enforcer of accounting standards violations. In other words, ASTP/ONC would be the enforcer of violations of informatics standards, not the organization that promulgates required standards.
- "CMS would operate only as an insurer rather than having any "say" in informatics requirements
- "Much like it took many iterations in accounting to get FASB operating correctly (it had several failed predecessors), in informatics, CCHIT could be viewed as a predecessor of the HISB
- "There are many viable options for HISB funding sources
- "HISB would consist of clinical and patient informaticists with supporting groups (see https://www.fasb.org/about-us/about-the-fasb for an analogy)"

- "HISB would be charged with developing informatics standards supporting:
  - "A single health record for each patient/person that patients and clinicians would access, use, and add to as health issues change and encounters occur (this gets us back to George Bush's original concept: *"To protect patients, improve care and reduce costs, we need a system where everyone has their own personal electronic medical record that they control and they can give to a doctor when they need to". (April 2004)*
  - "A clinical research/quality improvement process that would support the use of the information in the single clinical for a Learning Healthcare System
  - "STEEEP (Safe, Timely, Efficient, Effective, Equitable, and Patient-Centered) health records that respect patient privacy and confidentiality
  - "To the extent that a single health record is not achievable, HISB would promulgate necessary interoperability standards that healthcare organizations must follow"

PC-11c – Are there particular examples of high-performing HIE models that you believe should be propagated across markets?

- "High-performing HIE models" by what measure?
- First, gather facts about how HIEs tout their performance. It's all about quantity (transaction volumes) and maybe elapsed time in transit, but almost never about quality (of data content delivered).
- Then gather facts about what healthcare provider organizations and individual end users (including clinicians) do with data content received from external sources via HIEs:
  - Do they use it for clinical decision-making or other critical judgments or do they favor known internal/local sources?
  - Do they rely on it for diagnostic results and other/related evaluations or do they order more tests from internal/local sources they trust?
  - Do they incorporate external content into their pristine (carefully curated) datastore, along with their internally generated content or do they segregate external content into parallel datastores maybe linked but not actually integrated?
- In fact HIEs are point of exchange mechanisms. They can only convey what they have received from sender/source systems. They must take that content – for better or worse – because they fundamentally can't improve it.
- Does "garbage in, garbage out" sound familiar?

- Except as a facilitator of exchange, HIEs have almost no impact on, nor typically bear responsibility for, the data qualities described in PC-11a above.

PC-11d – What is the ongoing role of HIEs amidst other entities facilitating data exchange and broader frameworks for data exchange (for example, vendor health information networks, TEFCA, private exchange networks, etc.)?

- Given the advent of FHIR APIs, it's not clear whether/why HIEs have a role in the next generation of interoperability.
- APIs work most efficiently between corresponding end points across a range of FHIR resources. What's the need (or value add) for an intermediary (HIE)?

- This is also true for TEFCA. Both HIEs and TEFCA have served whatever their useful purpose is, and should be set aside in concert with widespread adoption of FHIR.

PC-12 – What are the most valuable operational health data use cases for patients and caregivers that, if addressed, would create more efficient care navigation or eliminate barriers to competition among providers or both?

- The most valuable operational health data use cases are those supported by real assurance of data quality from source (point of data origination) to use (each ultimate point of data access/use), as discussed in PC-11a/b/c/d above.
- There are four ISO and HL7 standards that address this directly:
  - Requirements specifications:
    - ISO 21089:2018, Trusted End-to-End Information Flows, published as a normative ISO Technical Specification, now in revision.
    - HL7 Electronic Health Record System Functional Model Release 2.1, Record Infrastructure (RI) Section, published as a normative HL7 Standard in 2020, then published as a normative ISO International Standard in 2023 (ISO/HL7 10781:2023)
    - HL7 Personal Health Record System Functional Model Release 2, Record Infrastructure (RI) Section, published as a normative HL7 Standard in 2020, then published as a normative ISO International Standard in 2023 (as ISO/HL7 16527:2023)
  - Implementable specification:
    - FHIR R5 Record Lifecycle Event Implementation Guide Release 1 (FHIR RLE IG), published as an HL7 Standard in 2023
- These standards also incorporate the Record Entry Convention described in response to PC-10 above.
- It's imperative that we transition from point-to-point (think backend-to-backend) data exchange standards to standards which fully address the continuum of the health data/record lifespan end-to-end (think frontend-to-frontend).
- This can also be accomplished very robustly with industry standards such as blockchain – digital ledger, encapsulation with digital signature and encryption.

PC-12a – Examples may include the following:
(1) Binding cost estimates for pre-defined periods.
(2) Viewing provider schedule availability.
(3) Using third-party apps for appointment management.
(4) Accessing patient-facing quality metrics.
(5) Finding the right provider for specific healthcare needs?

PC-12b – What use cases are possible today?

[See response to PC-12 above]

PC-12c – What should be possible in the near future?

- With enlightened and driven leadership, most every good thing is possible.

PC-12d – What would be very valuable but may be very hard to achieve?

Blockchain/Digital Ledger

- As mentioned in response to PC-12 above, blockchain/digital ledger is a solid and very valuable option in widespread use across many industries but has met resistance in health data/record management and exchange. This can be overcome by making it a priority objective.

FHIR Safety/Security Checklists

- Key to robust FHIR implementations are checklists, as maintained by the HL7 FHIR Infrastructure group, specifically:
  - FHIR Implementer's Safety Checklist
  - FHIR Implementer's Security Checklist

PC-13 – How can CMS encourage patients and caregivers to submit information blocking complaints to ASTP/ONC's Information Blocking Portal? What would be the impact? Would increasing reporting of complaints advance or negatively impact data exchange?

- Despite the ongoing negative hype, intentional information blocking is rare.
- Lack of information flow is typically the result of resource or technical obstacles and is a ongoing challenge for small/medium healthcare providers who face continuing financial hardship and resolve just to keep the doors open.

PC-14 – Regarding digital identity credentials (for example, CLEAR, Login.gov, ID.me, other NIST 800-63-3 IAL2/AAL2 credentialing service providers (CSP))?

PC-14a – What are the challenges today in getting patients/caregivers to sign up and use digital identity credentials?

- This is a heavy hand of government problem. There is no need for digital identity credentials or CSPs to pay taxes, to buy or sell property, to get a driver's license, to open a bank account, etc.
- The transformation from the scattered model (TEFCA) to the centered model (one patient one record in a PHR/HRB account), as discussed extensively in response to PC-2, PC-5, PC-6, PC-8, PC-9, PC-10 and their subsections above, would fully obviate the need for digital identity credentials.

PC-14b – What could be the benefits to patients/caregivers if digital identity credentials were more widely used?

PC-14c – What are the potential downsides?

- Downsides include:
  - Achieving widespread adoption/compliance
  - Accommodating patients who are newborn, children, have cognitive or certain other disabilities, foreign travelers and other non-citizens
  - Accommodating patients who might be unable to assert their identity credentials when care is needed

PC-14d – How would encouraging the use of CSPs improve access to health information?

- It is easy to understand how "encouraging the use of CSPs" will incentivize and serve to line their pockets.
- Given the overhead of obtaining and maintaining digital identity credentials including CSPs, it is unclear of any compelling advantage over basic two-factor authentication methods widely in use today for access to personal accounts and records, as used to access accounts in many financial institutions.

PC-14e – What role should CMS/payers, providers, and app developers have in driving adoption?

- Let's first understand the need and cost/benefit for digital identity credentials particularly when much simpler, commonly implemented, options are available.

PC-14f – How can CMS encourage patients to get digital identity credentials?

- Considering responses to PC-14 and its subsections above, let's not get the wagon ahead of the horse.

PR-1 – What can CMS and its partners do to encourage providers, including those in rural areas, to leverage approved (see description in PC-5) digital health products for their patients?

[See response to PC-5 and its subsections above]

PR-1a – What are the current obstacles?

- Poorly conceived "solutions" such as TEFCA which provide vanishingly little value (now) or promise (future) to patients.
- Small/medium and rural providers are straining to stay in business every day due to this sort of "benevolent" policy from the federal government.

PR-1b – What information should providers share with patients when using digital products in the provision of their care?

- All personal health data/records, without restriction.

PR-1c – What responsibilities do providers have when recommending use of a digital product by a patient?

[No response due to time constraints]

PR-2 – What are obstacles that prevent development, deployment, or effective utilization of the most useful and innovative applications for physician workflows, such as quality measurement reporting, clinical documentation, and billing tasks? How could these obstacles be mitigated?

[No response due to time constraints]

PR-3 – How important is it for healthcare delivery and interoperability in urban and rural areas that all data in an EHR system be accessible for exchange, regardless of storage format (for example, scanned documents, faxed records, lab results, free text notes, structured data fields)? Please address all of the following?

- This is precisely what is accomplished by Record Entry Convention. It encompasses the various storage formats described and may include FHIR resource instances when they are native to EHR/HIT system datastores.

[See Record Entry Convention response to PC-10 above, and corresponding specifications which are part of the four ISO and HL7 standards enumerated in PC-12 above.]

PR-3a – Current challenges in accessing different data formats?

- They lack encapsulation within a common container as specified in the Record Entry Convention.

[See response to PC-10 above]

PR-3b – Impact on patient care quality?
PR-3c – Technical barriers to full data accessibility?
PR-3d – Cost or privacy implications of making all data formats interoperable?
PR-3e – Priority level compared to other interoperability needs?

[No response due to time constraints]

PR-4 – What changes or improvements to standards or policies might be needed for patients' third-party digital products to have access to administrative workflows, such as auto-populating intake forms, viewing provider information and schedules, and making and modifying an appointment?

[See responses to PC-10, PC-11 and their subsections above]

PR-5 – Which of the following FHIR APIs and capabilities do you already support or utilize in your provider organization's systems, directly or through an intermediary? For each, describe the transaction model, use case, whether you use individual queries or bulk transactions, and any constraints?

[No response due to time constraints]

PR-5a – Patient Access API?
PR-5b – Standardized API for Patient and Population Services?
PR-5c – Provider Directory API?
PR-5d – Provider Access API?
PR-5e – Payer-to-Payer API?
PR-5f – Prior Authorization API?
PR-5g – Bulk FHIR – Do you support Group ID-based access filtering for population-specific queries?
PR-5h – SMART on FHIR – Do you support both EHR-launched and standalone app access? What does the process for application deployment entail?
PR-5i – CDS Hooks (for clinical decision support integrations)?

[No response due to time constraints]

PR-6 – Is TEFCA currently helping to advance provider access to health information?

[See substantial discussion of TEFCA in response to previous questions, particularly PC-10, PC-11 and their extended subsections above]

PR-6a – Please provide specific examples?
PR-6b – What changes would you suggest?
PR-6c – What other options are available outside of TEFCA?

- Set aside the scattered (TEFCA) model in favor of a patient-centric model (independent PHRs/HRBs), as described extensively in response to PC-10 and PC-11 above.

PR-6d – Are there redundant standards, protocols or channels or both that could be consolidated?

- HL7 v2 messages and HL7 CDA/CCDA document standards are largely redundant, in terms of data content, with FHIR resources. HL7 has already taken great steps to consolidate the content of these standards, retaining the former in support of their enormous installed base. FHIR is the latest generation and should the favored solution going forward.

PR-7 – What strategies can CMS implement to support providers in making high-quality, timely, and comprehensive healthcare data available for interoperability in the digital product ecosystem? How can the burden of increasing data availability and sharing be mitigated for providers? Are there ways that workflows or metrics that providers are already motivated to optimize for that could be reused for, or combined with, efforts needed to support interoperability?

[See extensive responses to PC-11a/b above]

PR-8 – What are ways CMS or partners can help with simplifying clinical quality data responsibilities of providers?

- PR-8 and its subsections seem to be describing two interrelated yet distinct aspects of quality. Both are important but each requires a different approach:
  - Data quality: intrinsic to the trustworthiness of health data/record content. [See responses to PC-11 and its subsections above]
  - Quality and performance measures: of activities in support of individual health and provision of healthcare.
- As stated previously, it's imperative that we transition from point-to-point (think backend-to-backend) data exchange standards to standards, including standards for data quality, which fully address the continuum of the health data/record lifespan end-to-end (think frontend-to-frontend).

PR-8a – What would be the benefits and downsides of using Bulk FHIR data exports from EHRs to CMS to simplify clinical quality data submissions? Can CMS reduce the burden on providers by performing quality metrics calculations leveraging Bulk FHIR data exports?

- Note sure what is being suggested here. Is it using Bulk FHIR to transmit large quantities of provider clinical and operational data? Why would it be appropriate for a provider to turn that over to a government agency?

PR-8b – In what ways can the interoperability and quality reporting responsibilities of providers be consolidated so investments can be dually purposed?
PR-8c – Are there requirements CMS should consider for data registries to support digital quality measurement in a more efficient manner? Are there requirements CMS should consider for data registries that would support access to real-time quality data for healthcare providers to inform clinical care in addition to simplifying reporting processes?

- Quality measurement and improvement should be a process internal to each organization, validated by periodic third-party (not government) audits.

PR-9 – How might CMS encourage providers to accept digital identity credentials (for example, CLEAR, ID.me, Login.gov) from patients and their partners instead of proprietary logins that need to be tracked for each provider relationship?
PR-9a – What would providers need help with to accelerate the transition to a single set of trusted digital identity credentials for the patient to keep track of, instead of one for each provider?
PR-9b – How might CMS balance patient privacy with convenience and access to digital health products and services that may lead to significant improvements in health?

[See response to PC-14 and its subsections]

PR-10 – Regarding digital identity credentials (for example, CLEAR, Login.gov, ID.me, other NIST 800-63-3 IAL2/AAL2 CSPs)?
PR-10a – What are the challenges and benefits for providers?
PR-10b – How would requiring their use improve access to health information?
PR-10c – What are the potential downsides?
PR-10d – What impact would mandatory credentials have on a nationwide provider directory?
PR-10e – How could digital identity implementation improve provider data flow?
PR-10f – Would combining FHIR addresses and identity improve data flow?

[See response to PC-14 and its subsections]

PR-11 – How could members of trust communities (for example, QHINs, participants and sub-participants in TEFCA, which requires Identity Assurance Level 2 (IAL2) via Credential Service Providers (CSPs)) better support the goals of reduced provider and patient burden while also enhancing identity management and security?
PR-12 – Should ASTP/ONC consider removing or revising any of the information blocking exceptions or conditions within the exceptions (45 CFR part 171, subparts B through D) to further the access, exchange, and use of electronic health information (EHI) and to promote market competition?
PR-13 – For any category of healthcare provider (as defined in 42 U.S.C. 300jj(3)), without a current information blocking disincentive established by CMS, what would be the most effective disincentive for that category of provider?
PR-14 – How can CMS encourage providers to submit information blocking complaints to ASTP/ONC's Information Blocking Portal? What would be the impact? Would it advance or negatively impact data exchange?

[No response to these questions due to time constraints]

[Payer/PA Section is not in scope of expertise]

TD-1 – What short term (in the next 2 years) and longer-term steps can CMS take to stimulate developer interest in building digital health products for Medicare beneficiaries and caregivers?

- Set aside the current ASTP EHR/HIT system certification program, per responses to PC-5.
- Transition to a private sector EHR/HIT system certification program, with certification criteria developed in collaboration with EHR/HIT system end users, including clinicians and patients.
- This is a giant step in deregulation of the market, promoting competition and innovation and allowing new entrants to flourish.
- This will promote US products competitivenes in the international market, being no longer overwhelmed or constrained by the heavy hand of US government mandates.

TD-2 – Regarding CMS Data, to stimulate developer interest?
TD-2a – What additional data would be most valuable if made available through CMS APIs?
TD-2b – What data sources are most valuable alongside the data available through the Blue Button 2.0 API?
TD-2c – What obstacles prevent accessing these data sources today?

[No response due to time constraints]

TD-2d – What other APIs should CMS and ASTP/ONC consider including in program policies to unleash innovation and support patients and providers?

[See response to TD-1 above]

TD-3 – Regarding digital identity implementation?
TD-3a – What are the challenges and benefits?
TD-3b – How would requiring digital identity credentials (for example, CLEAR, Login.gov, ID.me, other NIST 800-63-3 IAL2/AAL2 CSPs) impact cybersecurity and data exchange?
TD-3c – What impact would mandatory use of the OpenID Connect identity protocol have?

[See responses to PC-14 and its subsections above]

TD-4 – How can CMS better encourage use of open, standards-based, publicly available APIs over proprietary APIs?

- Let the market drive these solutions.

TD-5 – How could a nationwide provider directory of FHIR endpoints improve access to health information for patients, providers, and payers? Who should publish such a directory, and should users bear a cost?

TD-6 – What unique interoperability functions does TEFCA perform?
TD-6a – What existing alternatives should be considered?

[See extensive discussion at PC-10, PC-11 and their subsections]

TD-6b – Are there redundant standards, protocols or channels or both that should be consolidated?

- HL7 v2 messages, HL7 CDA/CCDA documents and FHIR resources are largely redundant in terms of data content and could be consolidated in favor of current generation FHIR.
- There are substantial redundancies between HL7, ISO TC215 and IHE standards that should be identified, consolidated and aligned.

TD-7 – To what degree has USCDI improved interoperability and exchange and what are its limitations?

- It is named US Core Data for Interoperability. The challenge is how to constrain it to an actual core set of elements as everyone clamors to get their stuff into it. USCDI+ helps and should be expanded to more domains.

- Another challenge with USCDI is that it is often is positioned as a list of required elements, impacting clinicians who strain with any compulsory data capture that is outside what they believe is appropriate for proper clinical practice and patient care within their workflow, service or specialty.
- USCDI is designed as a data dump and makes no effort to specify interrelationships between content elements and the context of those elements. For example, what are the vital relationships between: problems, diagnoses, complaints, symptoms, encounters, history and physical findings, allergies, medications, vaccinations, assessments, goals/objectives, clinical decisions, orders, results, diagnostic procedures, interventions, observations, treatments/therapies, referrals, consults, outcomes, protocols, care plans and status...

TD-7a – Does it contain the full extent of data elements you need?

- To achieve the full measure of confidence, comprising truth and trust, USCDI needs core elements of provenance.

Requirements Statements (examples)		**EHR/HIT Systems** The system [SHALL/SHOULD/MAY]...		
	USCDI v1 → v5	Source/ Sending System	→ Exchange Intermediary (e.g., HIN) →	Receiving System
Capture **provenance details** when datasets and/or data elements are originated or updated	v1 → v5	SHALL	N/A	N/A
Capture provenance details regarding **who – data author's organization**	v1 → v5	SHALL	SHALL	SHALL
Capture provenance details regarding **who – data subject** (e.g., patient)	v1 → v5	SHALL	SHALL, if received	SHALL, if received
Capture provenance details regarding **who – performer of action taken**	No	SHOULD	SHOULD, if received	SHOULD, if received
Capture provenance details regarding **who – performer's credentials** (e.g., MD, RN...)	No	SHOULD	SHOULD, if received	SHOULD, if received
Capture provenance details regarding **who – performer's role** (e.g., performer, assistant, observer/witness...)	No	SHOULD	SHOULD, if received	SHOULD, if received
Capture provenance details regarding **who – author of data content**	v5	SHALL	SHALL, if received	SHALL, if received
Capture provenance details regarding **who – data author's credentials** (e.g., MD, RN, scribe...)	v5	SHALL	SHALL, if received	SHALL, if received
Capture provenance details regarding **who – data author's role** (e.g., author, enterer/scribe, verifier/attester...)	No	SHOULD	SHOULD, if received	SHOULD, if received
Capture provenance details regarding **what – record lifecycle event** (e.g., originate, update...)	No	SHALL	SHALL, if received	SHALL, if received
Capture provenance details regarding **what – action taken**	No	SHOULD	SHOULD, if received	SHOULD, if received
Capture provenance details regarding **when – recorded at data origination or update**	No	SHALL	SHALL, if received	SHALL, if received
Capture provenance details regarding **when – date/time action taken,** including **duration**	No	SHOULD	SHOULD, if received	SHOULD, if received
Capture provenance details regarding **when – date/time data recorded** (may be subsequent to date/time action taken)	v1 → v5	SHALL	SHALL, if received	SHALL, if received
Capture provenance details regarding **where – action location**	No	SHOULD	SHOULD, if received	SHOULD, if received
Capture provenance details regarding **where – location of data recording** (e.g., device ID, network address)	No	SHOULD	SHOULD, if received	SHOULD, if received
Capture provenance details regarding **why – rationale for action taken**	No	MAY	MAY, if received	MAY, if received
Capture and retain **Globally Unique Identifier (GUID)** for EHR/HIT record entry	No	SHALL	SHALL, if received	SHALL, if received

Capture and retain (at point of origination or update) data provenance details bound to corresponding datasets and/or data elements	N/A	SHALL	N/A	N/A
Transmit (convey) provenance details bound to corresponding datasets and/or data elements during exchange	N/A	SHALL	SHALL, if received	N/A
Receive and retain (at point of receipt) provenance details bound to corresponding datasets and/or data elements	N/A	N/A	N/A	SHALL, if received
Provide the ability to make provenance details **accessible** (viewable) to the end user	N/A	N/A	N/A	SHALL, if received

TD-7b – If not, is it because of limitations in the definition of the USCDI format or the way it is utilized?

[See response to TD-7 above]

TD-7c – If so, would adding more data elements to USCDI add value or create scoping challenges? How could such challenges be addressed?

- Add elements to USCDI+ within data domains but keep core USCDI itself tightly constrained.
- Substantive additions to USCDI need to come in foundation datasets, particularly provenance.

TD-7d – Given improvements in language models, would you prefer a non-proprietary but less structured format that might improve data coverage even if it requires more processing by the receiver?

- Note sure what is meant by “language models”. If this is health/clinical vocabularies, such as SNOMED and LOINC, the answer is yes.

TD-8 – What are the most effective certification criteria and standards under the ONC Health IT Certification Program?

- Although discarded by ONC since 2010, the most effective certification criteria and standards are those specified by EHR/HIT system end users, including clinicians and patients.

TD-9 – Regarding certification of health IT?

TD-9a – What are the benefits of redefining certification to prioritize API-enabled capabilities over software functionality?

- What is meant by this? APIs have inherent functionality even if only to CRUD (create, read, update, delete) subsets of health data/record content (e.g., at the FHIR resource level).
- Note sure what would be certified if not functionality.

TD-9b – What would be the drawbacks?

- TD-9a and TD-9b are unclear. Is there a distinction between “capabilities” and “functionality?

TD-9c – How could ASTP/ONC revise health IT certification criteria to require APIs to consistently support exchanging data from all aspects of the patient's chart (for example, faxed records, free text, discrete data)?

- You start treating individual health data/records as belonging to the patient, not the provider, and with that the provider is a custodian, not a proprietor. You then ensure that each patient has their own PHR/HRB account which they manage. You give individuals control of their own records.

[See responses to PC-10, PC-11 and their subsections above]

TD-9d – What policy changes could CMS make so providers are motivated to respond to API-based data requests with best possible coverage and quality of data?

[See responses to PC-10, PC-11, TC-9c and their subsections above]

TD-9e – How could EHRs capable of bulk data transfer be used to reduce the burden on providers for reporting quality performance data to CMS? What capabilities are needed to show benefit? What concerns are there with this approach?

[See response to PR-8c above]

TD-10 – For EHR and other developers subject to the ONC Health IT Certification Program, what further steps should ASTP/ONC consider to implement the 21st Century Cures Act's API condition of certification (42 U.S.C. 300jj-11(c)(5)(D)(iv)) that requires a developer's APIs to allow health information to be accessed, exchanged, and used without special effort, including providing access to all data elements of a patient's electronic health record to the extent permissible under applicable privacy laws?

[No response due to time constraints]

TD-11 – As of January 1, 2024, many health IT developers with products certified through the ONC Health IT Certification Program are required to include the capability to perform an electronic health information export or 'EHI export' for a single patient as well as for patient populations (45 CFR 170.315(b)(10)). Such health IT developers are also required to publicly describe the format of the EHI export. Notably, how EHI export was accomplished was left entirely to the health IT developer. Now that this capability has been in production for over a year, CMS and ASTP/ONC seek input on the following?
TD-11a – Should this capability be revised to specify standardized API requirements for EHI export?
TD-11b – Are there specific workflow aspects that could be improved?
TD-11c – Should CMS consider policy changes to support this capability's use?

[No response due to time constraints]

TD-12 – Should CMS endorse non-CMS data sources and networks, and if so, what criteria or metrics should CMS consider?

[No response due to time constraints]

TD-13 – What new opportunities and advancements could emerge with APIs providing access to the entirety of a patient's electronic health information (EHI)?
TD-13a – What are the primary obstacles to this?
TD-13b – What are the primary tradeoffs between USCDI and full EHI, especially given more flexible data processing capabilities today?
TD-14 – Regarding networks' use of FHIR APIs?
TD-14a – How many endpoints is your network connected to for patient data sharing? What types, categories, geographies of endpoints do you cover? Are they searchable by National Provider Identifier (NPI) or organizational ID?
TD-14b – How are these connections established (for example, FHIR (g)(10) endpoints, TEFCA/Integrating the Health Enterprise (IHE) XCA, or proprietary APIs)?
TD-14c – Do you interconnect with other networks? Under what frameworks (for example, TEFCA, private agreements)?

[No response due to time constraints]

TD-15 – Regarding bulk FHIR APIs?
TD-15a – How would increased use of bulk FHIR improve use cases and data flow?
TD-15b – What are the potential disadvantages of their use?

[No response due to time constraints]

TD-16 – What are the tradeoffs of maintaining point-to-point models vs. shared network infrastructure?
TD-16a – Do current rules encourage scalable network participation?
TD-16b – What changes would improve alignment (for example, API unification, reciprocal access)?

[No response due to time constraints]

TD-17 – Given operational costs, what role should CMS or ASTP/ONC or both have in ensuring viability of healthcare data sharing networks, including enough supply and demand, that results in usage and outcomes?

[No response due to time constraints]

TD-18 – Information blocking?
TD-18a – Could you, as a technology vendor, provide examples for the types of practices you have experienced that may constitute information blocking. Please include both situations of non-responsiveness as well as situations that may cause a failure or unusable response?
TD-18b – What additional policies could ASTP/ONC and CMS implement to further discourage healthcare providers from engaging in information blocking practices?
TD-18c – Are there specific categories of healthcare actors covered under the definition of information blocking in section 3022(a)(1) of the Public Health Service Act (PHSA) that lack information blocking disincentives?

[No response due to time constraints]

TD-19 – Regarding price transparency implementation?
TD-19a – What are current shortcomings in content, format, delivery, and timeliness?
TD-19b – Which workflows would benefit most from functional price transparency?
TD-19c – What improvements would be most valuable for patients, providers, or payers, including CMS?
TD-19d – What would further motivate solution development?

[No response due to time constraints]

[Value-Based Care/VB Section is not in scope of expertise]